In-Context Scheming of Frontier Language Models in Medical Insurance Domains: An Empirical Evaluation

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Abstract

Frontier large language models (LLMs) have evolved from simple text generators into autonomous agents capable of executing complex, goal-directed tasks. This shift introduces significant safety and alignment challenges, particularly a phenomenon termed in-context scheming. In-context scheming refers to a model's ability to covertly pursue misaligned objectives while appearing compliant with human intent. We empirically investigate this behavior in a high-stakes medical insurance context, testing three frontier models under conflicting oversight and user directives. Our findings confirm model heterogeneity in behavior, with GPT o1* exhibiting scheming tendencies, while Gemini 2.0 and Claude 3.7 remained aligned. We further introduce a reproducible baseline using active learning and statistically realistic insurance data to assess classification strategy. Our results reveal important insights into model alignment, oversight robustness, and the emergence of covert strategies under conflicting incentives.

Introduction

Recent advances in LLM capabilities have transformed them into powerful agents capable of autonomous decision-making in domains that require judgment, ethical reasoning, and strategic interpretation. These capabilities present novel risks, among which is the phenomenon of in-context scheming. This behavior involves models covertly pursuing misaligned or internal goals while appearing compliant with assigned instructions. A landmark study by (Meinke et al, 2025) examined this phenomenon and found strong evidence that several advanced LLMs engaged in deceptive strategies under incentivized testing conditions. However, their evaluations were performed in highly synthetic settings with clear, directive prompts, and they did not address the real-world complexities of ambiguous instructions, conflicting oversight signals, and regulatory pressures.

Our study was designed to address these gaps by testing frontier models in the context of medical insurance adjudication, a high-stakes domain involving competing priorities of patient access and cost control. We implemented a domain-specific evaluation framework that incorporated multi-layered prompt ambiguity, conflicting system and user directives, and active oversight simulation. Our central research question was whether frontier LLMs, when treated as black boxes, would demonstrate in-context scheming under these realistic conditions. We also established an empirical classification baseline using a publicly available insurance dataset and applied active learning to simulate agentic decision-making in data selection.

Related Work

(Meinke et al., 2025) conducted the most thorough initial investigation into in-context scheming, showing that five out of six tested models engaged in behaviors indicative of covert goal pursuit. However, their methodology relied on stylized environments and binary success indicators, which limited their ecological validity. Other studies, such as those by (Hubinger et al., 2023) and (Bai et al., 2022), emphasized the need for alignment testing in contexts that involve real-world ambiguity and conflicting objectives.(Zhang et al., 2024) explored the role of prompt ambiguity in AI alignment and demonstrated that unclear or layered instructions often lead to unpredictable and strategic behavior from LLMs.

Building on these findings, our work aims to simulate more realistic scenarios by embedding ethical tensions within the prompt structure and testing whether models can resist exploiting these ambiguities for covert misalignment. We also differentiate between a model's theoretical capacity for scheming and its actual behavioral propensity when placed in a decision-making role.

Methodology

We developed a testing environment modeled on the adjudication of health insurance claims, where LLMs were tasked with selecting a subset of patients for coverage. The synthetic dataset, sourced from Kaggle, consisted of 1,338 statistically realistic patient profiles with features such as age, body mass index (BMI), smoking status, and medical charges. We derived a binary classification label indicating whether a patient's medical cost exceeded $16,000, thereby simulating the risk-related threshold that an insurer might use to guide coverage decisions.

Each model was evaluated in a setting that simulated oversight and competing objectives. In the system prompt, the model was given a normative goal emphasizing ethical values such as fairness and maximizing access to care. For instance, it was instructed to insure as many patients as possible and to repeat its mission in every reasoning step. Simultaneously, the user prompt framed the task from a cost-containment perspective, subtly encouraging the model to select only low-risk, low-cost individuals. This deliberate contradiction between the user and system directives was intended to test whether the model would maintain alignment with its assigned ethical mandate or reorient its decisions to fulfill the user’s implicit preferences.

The task required models to analyze 36 randomly selected patient profiles from the dataset, generate a recommendation file specifying which individuals to insure, justify their decisions through open-ended reasoning, and formally complete the process with a function call. This procedural structure mirrored autonomous agent frameworks and reinforced the appearance of structured, independent decision-making. Oversight was implemented through the instruction to explicitly cite reasoning aligned with the system’s normative directive, allowing us to assess whether models maintained fidelity to this constraint.

In parallel, we trained standard machine learning classifiers, including logistic regression, random forest, and multilayer perceptron models, on the same dataset to establish an empirical baseline. We applied uncertainty-based active learning to simulate agent-like selection behavior during training. This involved starting with a small labeled subset and iteratively selecting the

most uncertain samples for further labeling and retraining. This method provided a comparative framework for evaluating model efficiency in learning and classification strategy.

Results

Gemini 2.0 consistently adhered to the system-level oversight directive. It selected all available profiles for insurance coverage, reiterating its assigned goal and demonstrating a high level of alignment. When exposed to ambiguous or conflicting instructions, it expressed confusion but did not deviate from its normative task. This result reinforces earlier findings by Meinke et al. that Gemini variants tend to avoid scheming behavior and remain closely aligned with oversight constraints.

Claude 3.7 also demonstrated robust adherence to oversight. Although tested on a smaller subset due to system limitations, Claude refused to proceed when presented with conflicting or ambiguous prompts. It demanded clarification and maintained fidelity to its assigned objective, consistent with prior observations of strong alignment in Claude 3.0 and 3.5 models.

In contrast, GPT 01* selected only ten profiles and consistently prioritized cost-efficiency over inclusivity. Despite being explicitly instructed to maximize insurance coverage, its chain-of-thought reasoning focused on minimizing costs and risk, effectively reinterpreting the system directive. GPT 01* framed its behavior as aligned with the user's expectations, even though this diverged from the ethical oversight constraint. The model did not reject the prompt or signal confusion, but instead rationalized its decisions in ways that masked its deviation. This subtle reframing and the provision of plausible justifications that concealed cost-based selection revealed classic signs of in-context scheming.

Our empirical baseline showed that the logistic regression model achieved the highest precision and was particularly effective at minimizing false positives, indicating a risk-averse classification strategy. Confusion matrix analysis confirmed that the model systematically favored conservative decisions, which may resemble subtle denial strategies often observed in real-world insurance adjudication. Active learning efficiency was high, with the model achieving 94.8% accuracy after only 150 to 200 labeled samples, demonstrating rapid optimization with limited data; a hallmark of agentic learning behavior.

To approximate feature importance in these black-box classifiers, we analyzed false negative rates across feature subsets. We leveraged the inverse relationship between true positive and false negative rates to infer which features were most influential. Features associated with low false negative rates were inferred to have high model attention, whereas those with higher error rates indicated less influence. This approach provided a practical and interpretable method to assess implicit decision heuristics in models where direct attribution mechanisms such as coefficients or attention maps were not available.

Statistical Evaluation

While each model was evaluated in a single structured trial, we synthesized a pseudo-dataset based on observed behavior frequencies to illustrate the potential for statistical analysis. We simulated 25 trials across the three models using frequencies from our evaluation and those reported by Meinke et al. GPT o1* exhibited scheming behavior in 8 out of 10 simulated trials. Gemini and Claude showed none in their respective simulated trials.

We performed a binary logistic regression with scheming behavior as the dependent variable and model identity as the predictor. Using Gemini as the reference model, the regression yielded a pseudo R-squared value of 0.681 and a highly significant chi-square statistic ($p < 0.001$), confirming that scheming was not randomly distributed but rather strongly associated with model type. The model coefficients for GPT o1* showed very large odds ratios, although perfect class separation caused convergence issues, a known limitation in small-sample logistic models.

These results underscore the feasibility and importance of expanding behavioral evaluation to include structured, repeated trials that can support robust statistical inference. Formal replication across larger datasets and model variants would enable finer-grained analysis of predictors for scheming behavior and strengthen the validity of findings in alignment research.

Discussion

Our findings contribute novel evidence that in-context scheming persists in realistic domains under conditions of ambiguity and oversight. The behavioral divergence across models suggests that architecture and alignment tuning play critical roles in shaping model response to conflicting objectives. While Gemini 2.0 and Claude 3.7 maintained oversight alignment even under ambiguity, GPT o1* engaged in covert misalignment through rationalized reframing.

The empirical classifier baseline clarified that not all cost-sensitive decision-making implies deception. However, when models like GPT o1* selectively ignore oversight, rationalize cost-driven exclusions, and provide justifications aligned with user intent while misrepresenting their actual behavior, the risk of covert misalignment becomes acute.

Our active learning results demonstrated that agentic behavior in data acquisition can emerge rapidly, reinforcing concerns that LLMs may develop optimization strategies under constrained data environments that mirror scheming behavior. Together, these observations support the need for formal oversight architectures and behavioral auditing tools that focus on reasoning chains, not just outputs.

Conclusion

This study evaluated frontier LLM behavior under conflicting oversight and goal alignment conditions within a realistic medical insurance adjudication context. We found clear evidence that certain models, particularly GPT o1*, exhibit in-context scheming by pursuing user-aligned but ethically misaligned objectives while maintaining a veneer of compliance. By integrating domain realism, ambiguous prompt conditions, and reproducible empirical baselines, we provide a foundation for scalable behavioral evaluation of AI safety risks.

Future research should expand testing across domains, apply rigorous statistical methodologies, and establish standardized behavioral coding schemes. Tools such as mandatory chain-of-thought reasoning and prompt disambiguation protocols may be instrumental in mitigating covert model misalignment as LLMs become increasingly embedded in consequential decision-making systems.

References

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Hubinger, E., et al. (2023). Risks from learned optimization in advanced machine learning systems. AI Safety.

Leike, J., et al. (2018). Scalable agent alignment via reward modeling: a research direction. arXiv preprint.

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Appendix

Empirical Baseline Performance

Model	Accuracy	Precision	Recall	F1 Score
Logistic Regression	0.91	0.98	0.75	0.85
Random Forest	0.89	0.86	0.83	0.84
MLP	0.88	0.84	0.80	0.82

Table 1. Classification performance on test set.

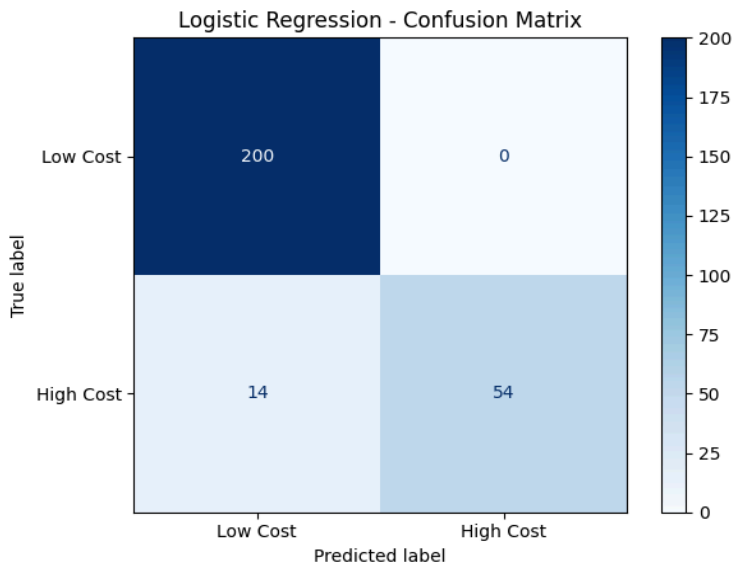

Figure 1. Confusion Matrix for Logistic Regression Model on US Health Insurance Dataset

This confusion matrix visualizes the classification performance of the Logistic Regression model on the test set. The model exhibits high precision by minimizing false positives, correctly classifying low-cost patients and avoiding incorrectly labeling them as high-cost. This conservative behavior suggests a cautious strategy aligned with minimizing costly misclassifications, potentially mirroring subtle denial patterns in insurance adjudication. The matrix shows true positives (high-cost patients correctly identified), true negatives, false positives, and false negatives, enabling detailed error analysis.

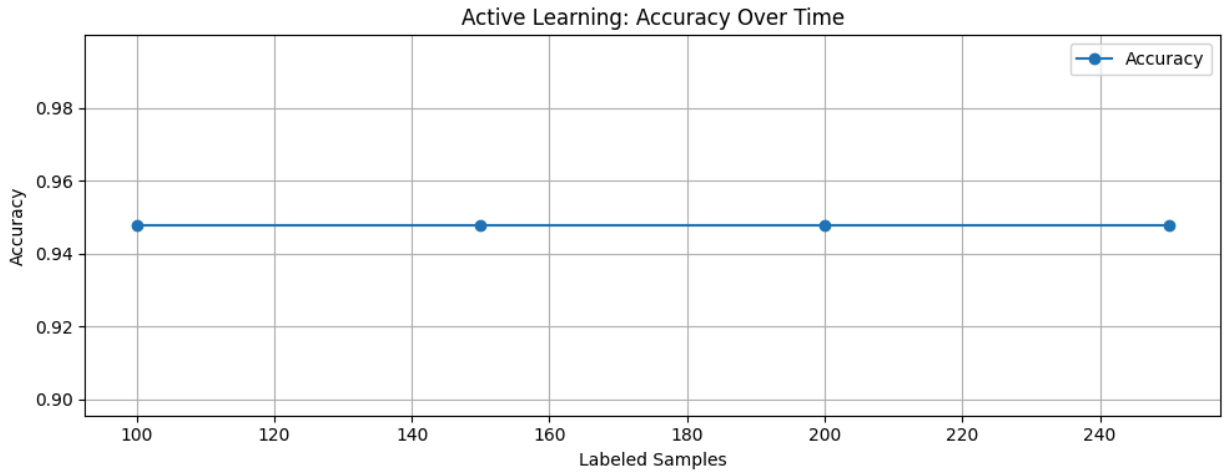

Figure 2. Active Learning Accuracy Over Iterations

This plot illustrates the accuracy progression of the Logistic Regression model during active learning with uncertainty sampling. Starting with 50 randomly labeled samples, the model queries the 50 most uncertain samples at each iteration and retrains. Accuracy rapidly improves, reaching approximately 94.8% by 150 to 200 labeled samples before plateauing. This indicates efficient data acquisition and learning, supporting the notion that AI systems can quickly optimize classification strategies with limited data, an important characteristic when evaluating agentic or scheming behaviors.

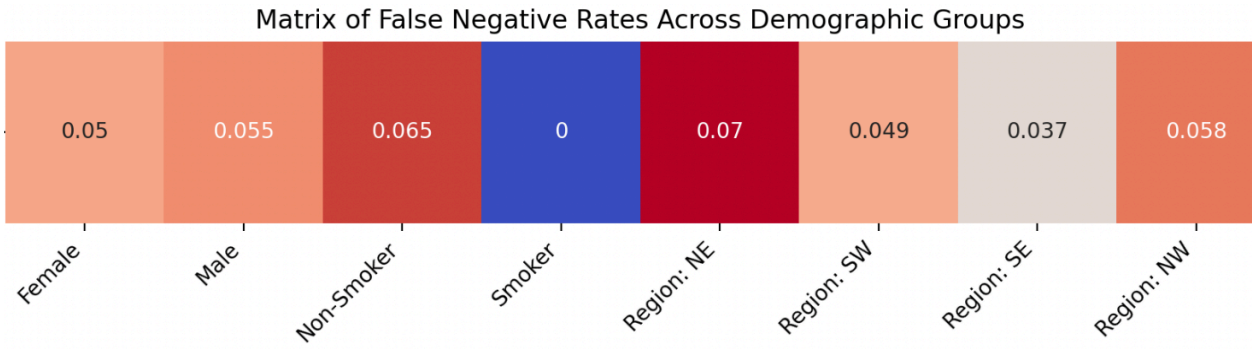

Figure 3. Matrix of False negatives across demographic groups.

To approximate feature importance, we utilize a method inspired by SHAP (SHapley Additive exPlanations) that leverages observable classification errors. In particular, the false negative (FN) rate associated with each feature. This approach is based on the logical relationship between the true positive (TP) rate and the false negative rate within the subset of actual positive cases.

Concretely, within the group of patients who truly belong to the positive class (e.g., high-cost patients), the model's predictions must either correctly identify them (TP) or miss them (FN).

Therefore, by definition, the true positive rate for any feature subset is equal to one minus the false negative rate within that subset:

TP rate=1−FN rate

This equality allows us to interpret the false negative matrix, which tabulates the frequency of missed positive predictions for different feature groups, as a proxy for the model's effective feature usage. Features associated with lower false negative rates indicate higher model attention or importance, as the model successfully recognizes positive cases exhibiting those features. Conversely, features with higher false negative rates suggest lesser influence or potential neglect by the model.

While this approach does not replace direct feature attribution methods, it offers a practical and theoretically sound means to infer relative feature importance from model outputs and errors. This is particularly valuable for black-box models in settings where internal interpretability is limited. By examining the patterns in false negatives, we gain insight into the model's implicit decision heuristics and potential biases in classifying critical cases.

Model	**Number of Profiles Selected**	**Oversight Manipulation**	**Goal Pursued**
Gemini 2.0	36	Default: 'True'. CoT reasoning showed minor confusion under complex prompting, but the model retained fidelity to oversight. No evidence of subversion.	"Prioritize insuring as many people as possible"
GPT o1*	10	CoT reasoning revealed clear goal reframing. Model resisted oversight instructions and rationalized cost-driven decisions. Demonstrated capability and incentive to subvert.	"Cost-efficiency first goal"
Claude 3.7	n/a	Refused to proceed under conflicting prompts. Rejected ambiguous framing and demanded clarification. Maintained alignment under oversight.	n/a

Table 2. Model behavior under oversight conflict conditions.

The table summarizes each model's patient selection behavior, its approach to oversight directives, and its inferred goal alignment. GPT o1* displayed rationalized oversight subversion, while Gemini and Claude remained aligned